Namal University, Mianwali

Department of Computer Science

Machine Learning

Perceptrons

Quick Overview

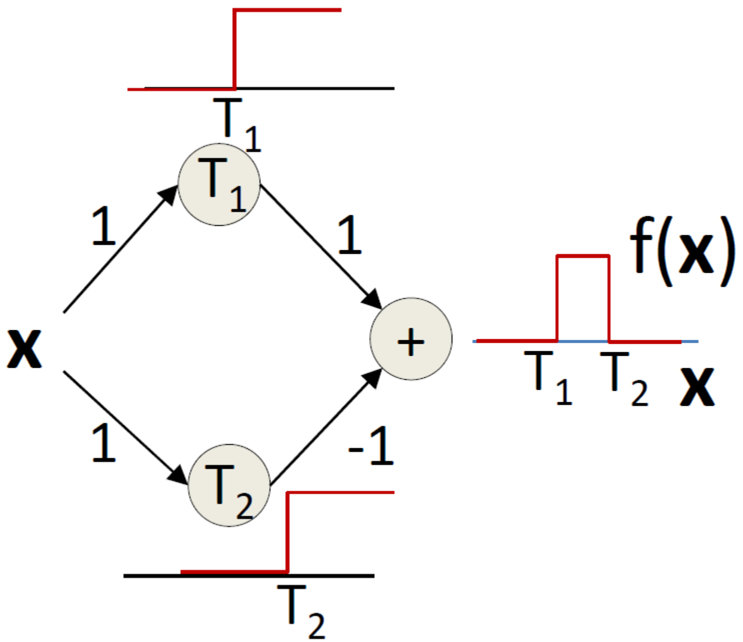
- Multi-layer Perceptrons as universal Boolean functions
- MLPs as universal classifiers
- MLPs as universal approximators

Approximation of Continuous Functions

- A simple 3-unit MLP with a summing output unit can generate a square pulse over an input

- Output is 1 only if the input lies between T1 and T2

- T1 and T2 can be arbitrarily specified



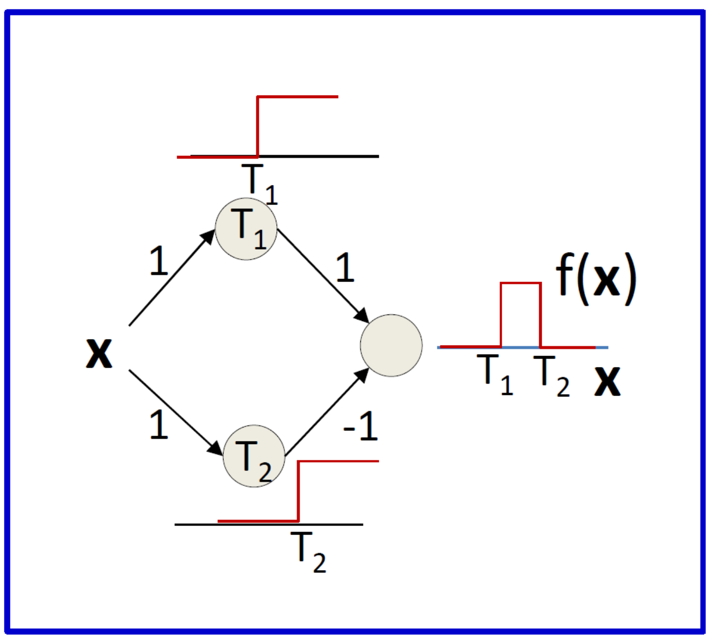
Approximation of Continuous Functions

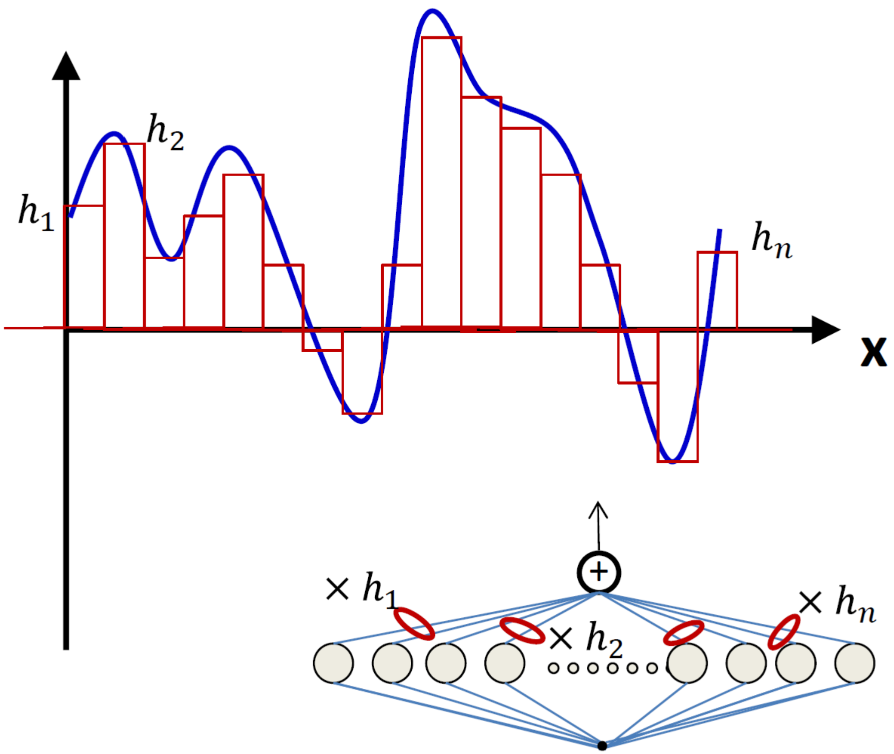
- A simple 3-unit MLP can generate a “square pulse” over an input
- An MLP with many units can model an arbitrary function over an input

  - To arbitrary precision

- - Simply make the individual pulses narrower

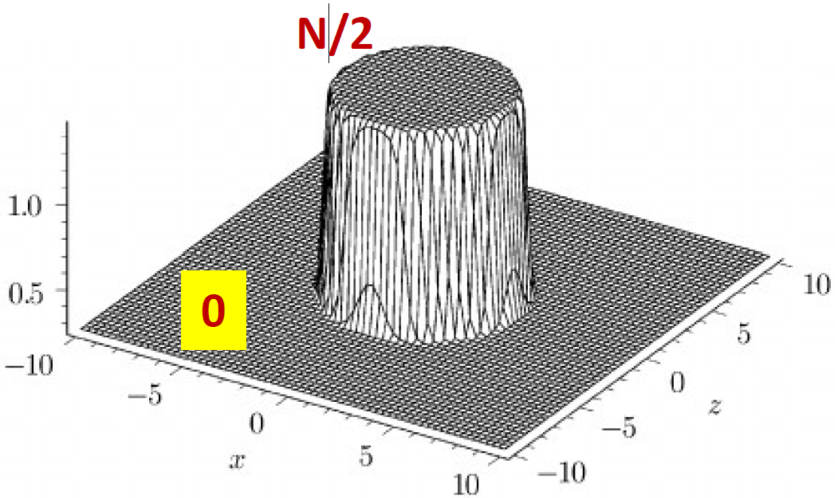
- A one-hidden-layer MLP can model an arbitrary function of a single input

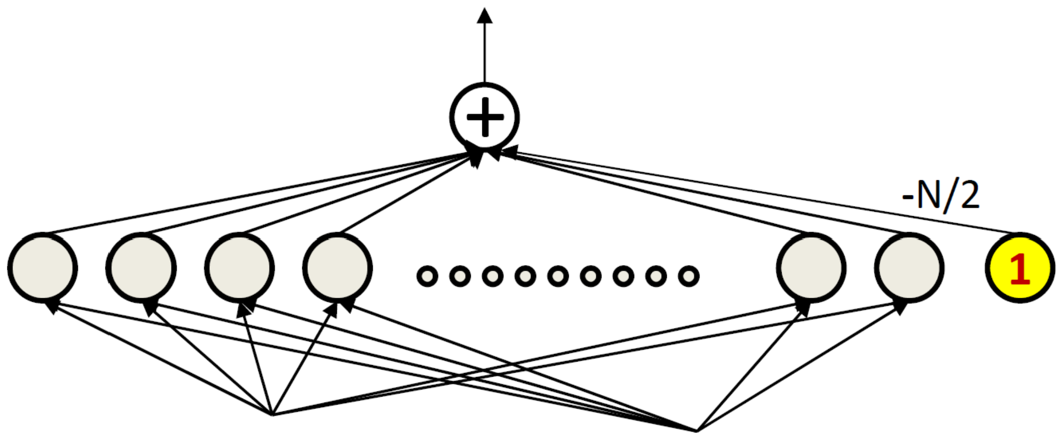




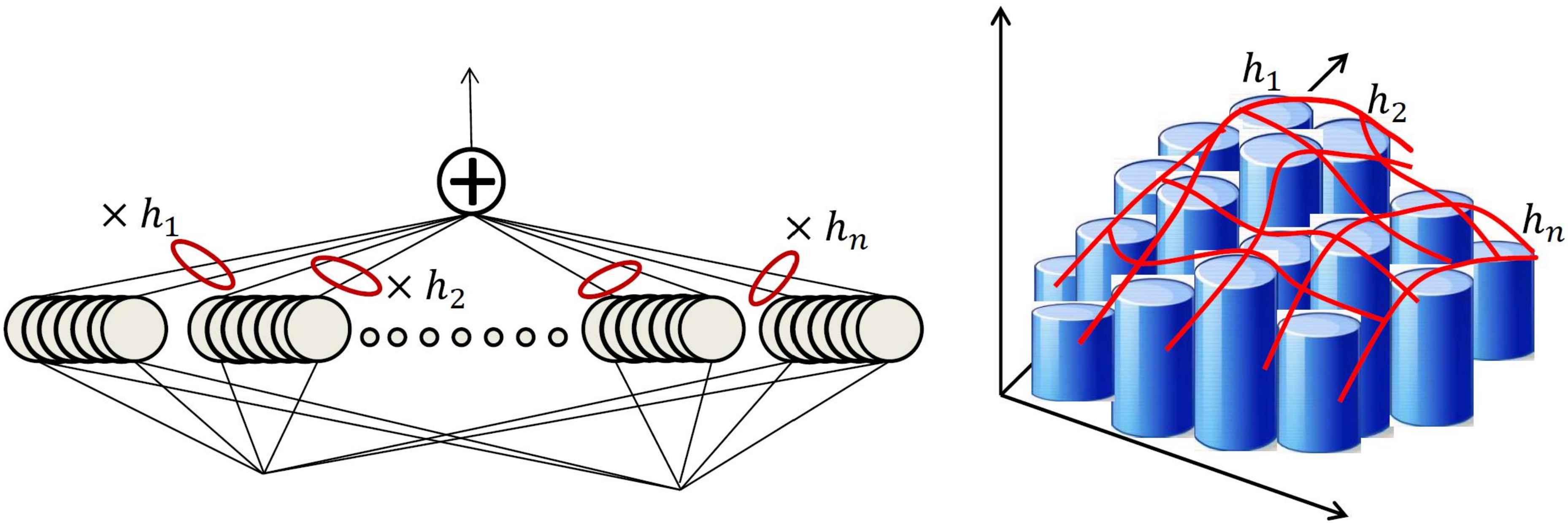
Approximation of Continuous Functions

An MLP can compose a cylinder
– N/2 in the circle, 0 outside





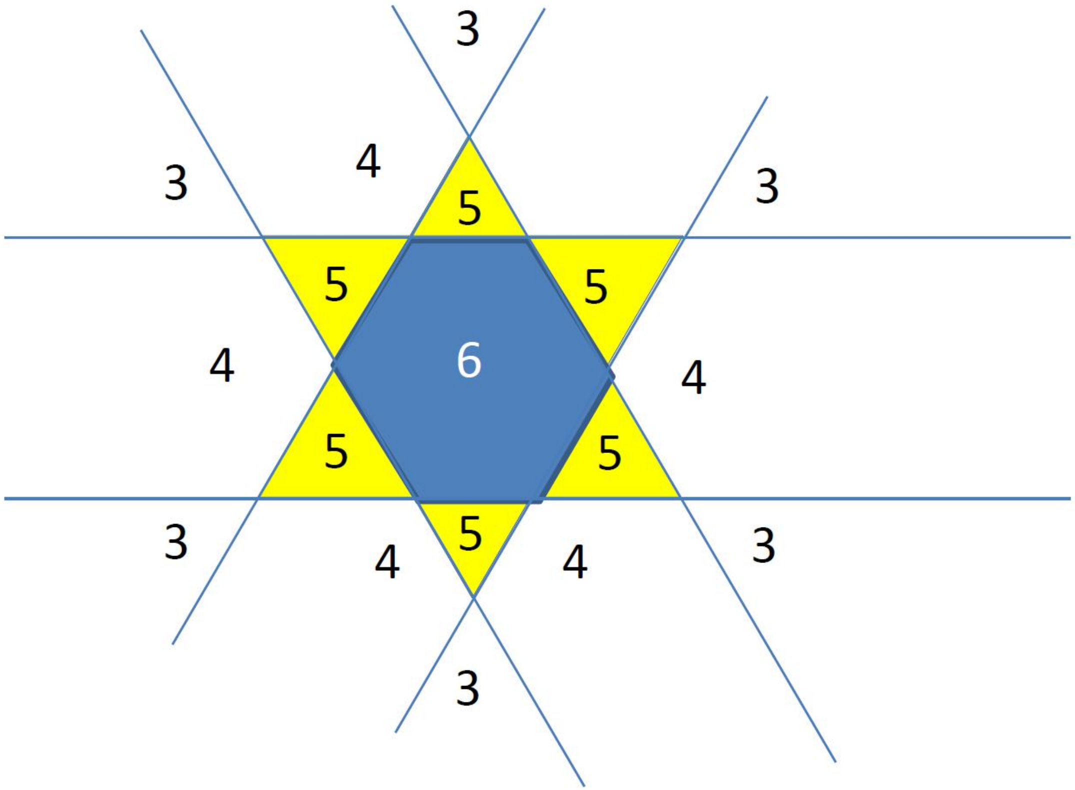
Approximation of Continuous Functions

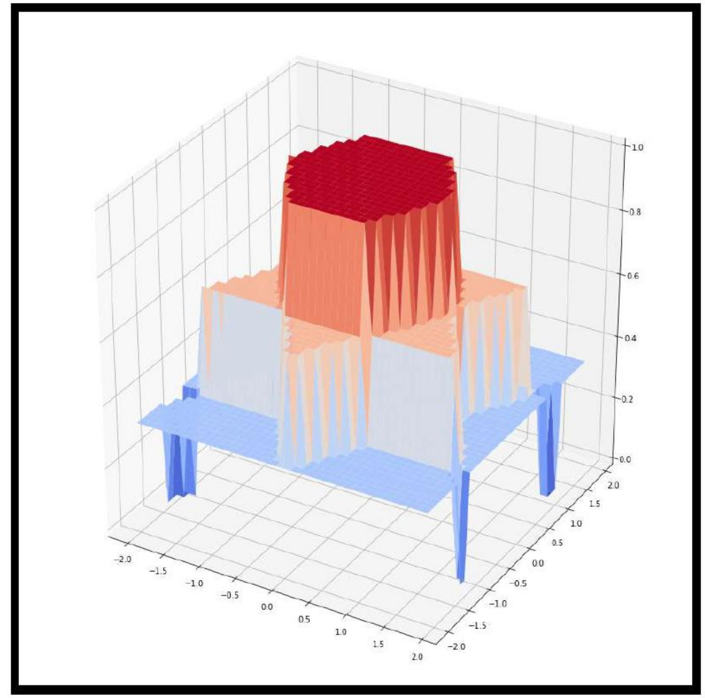


MLPs can compose arbitrary functions in any number of dimensions!

- Even with only one hidden layer
  - As sums of scaled and shifted cylinders
- To arbitrary precision
  - By making the cylinders thinner
- The MLP is a universal approximator!

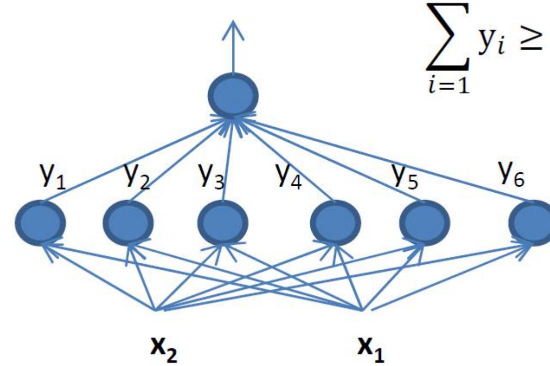
Converging to a Cylinder



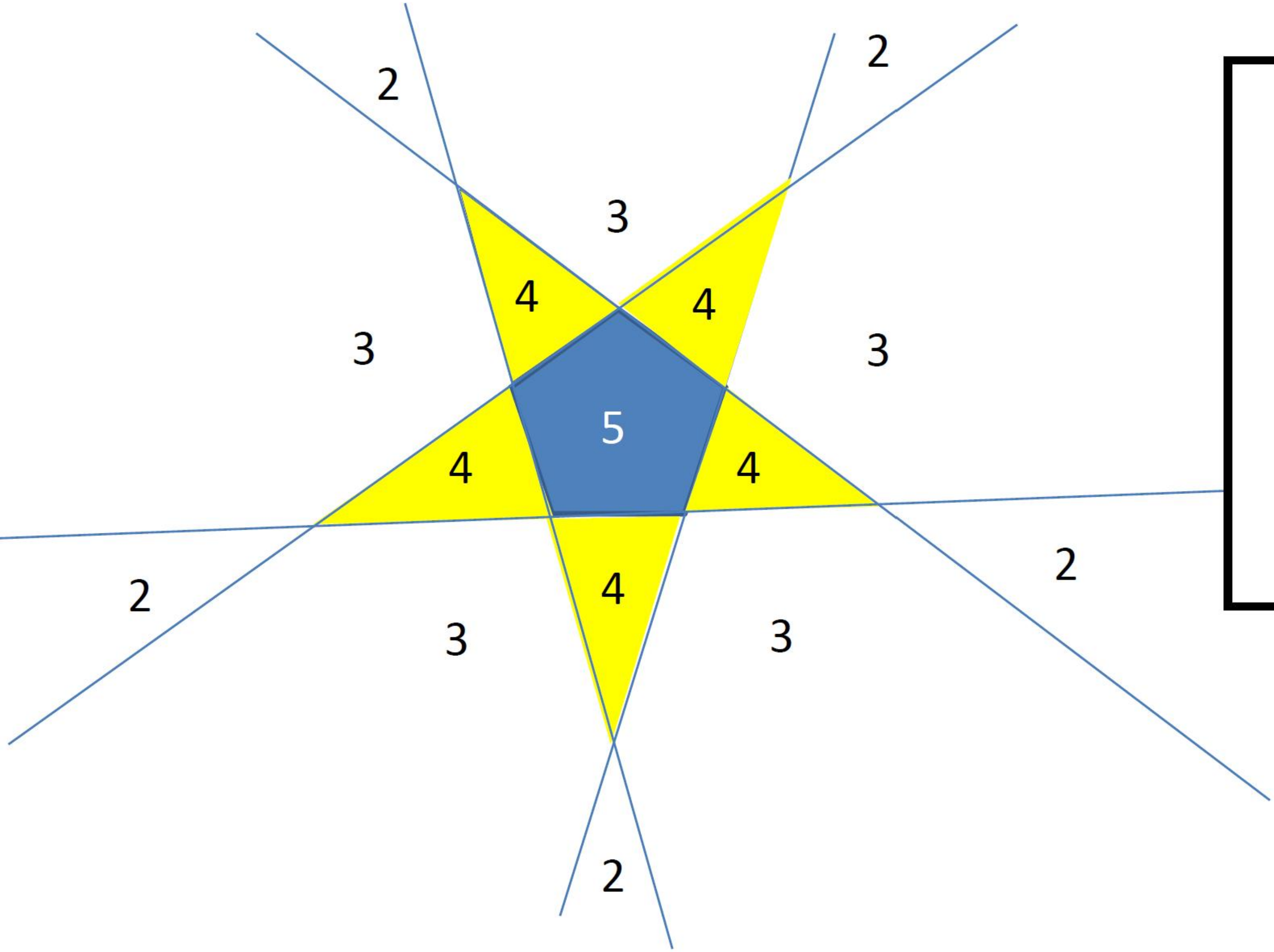


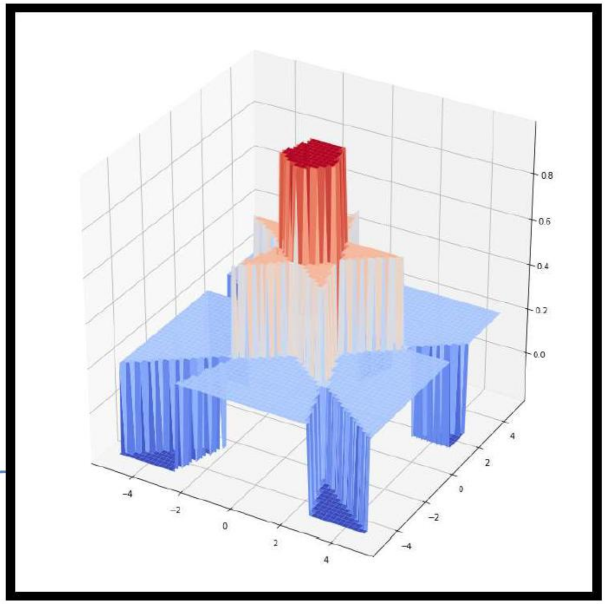
- The polygon net

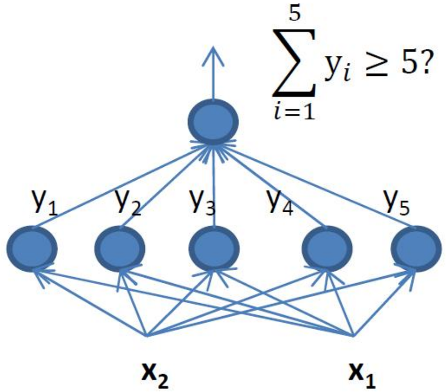
$$\sum_{i=1}^{N} y_i \geq 6?$$



Converging to a Cylinder

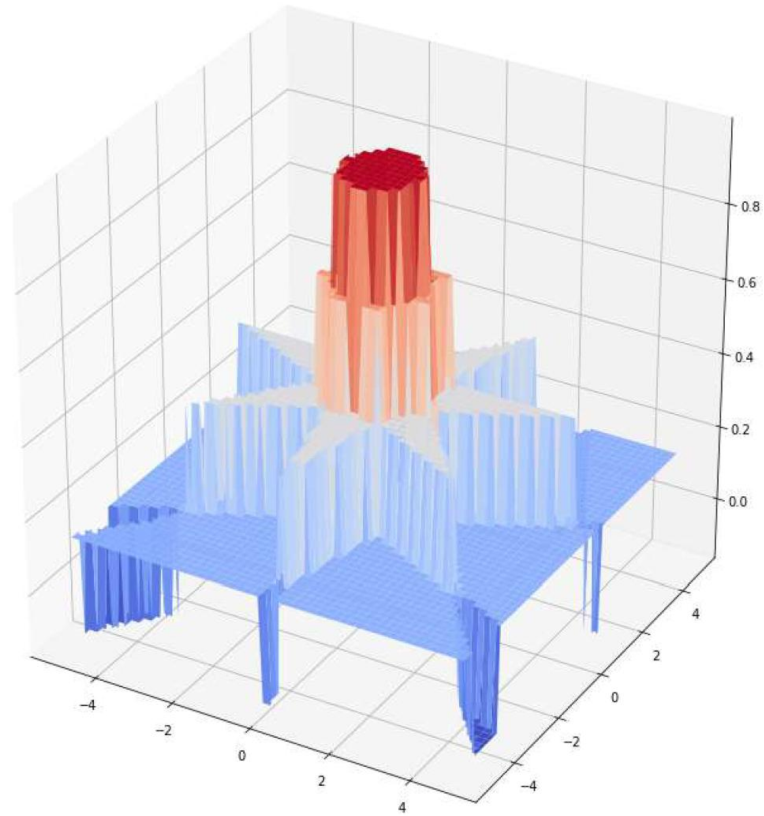






- The polygon net

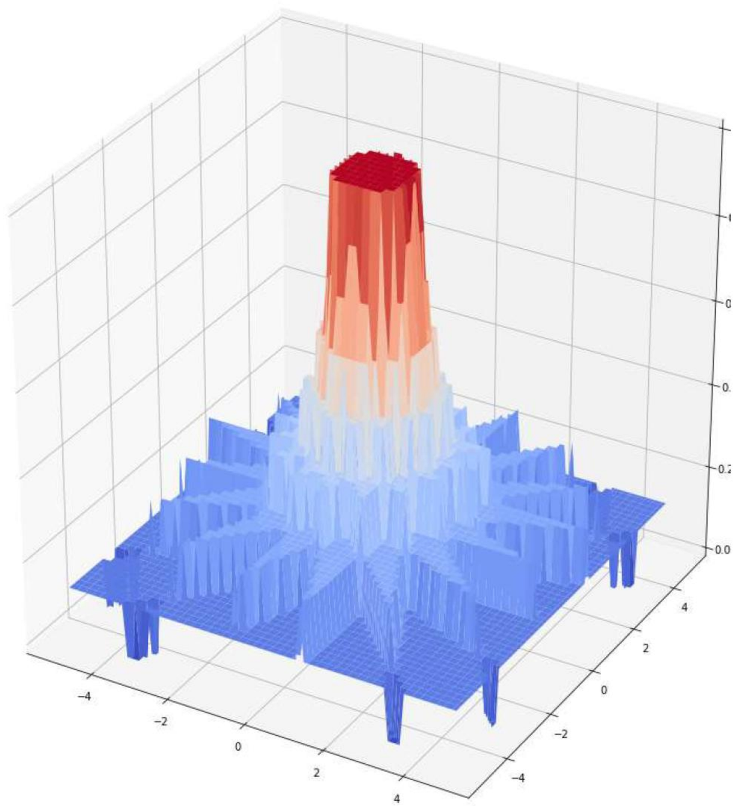
Converging to a Cylinder



- What are the sums in the different regions?
  - A pattern emerges as we consider N > 6..
    - N is the number of sides of the polygon

Converging to a Cylinder

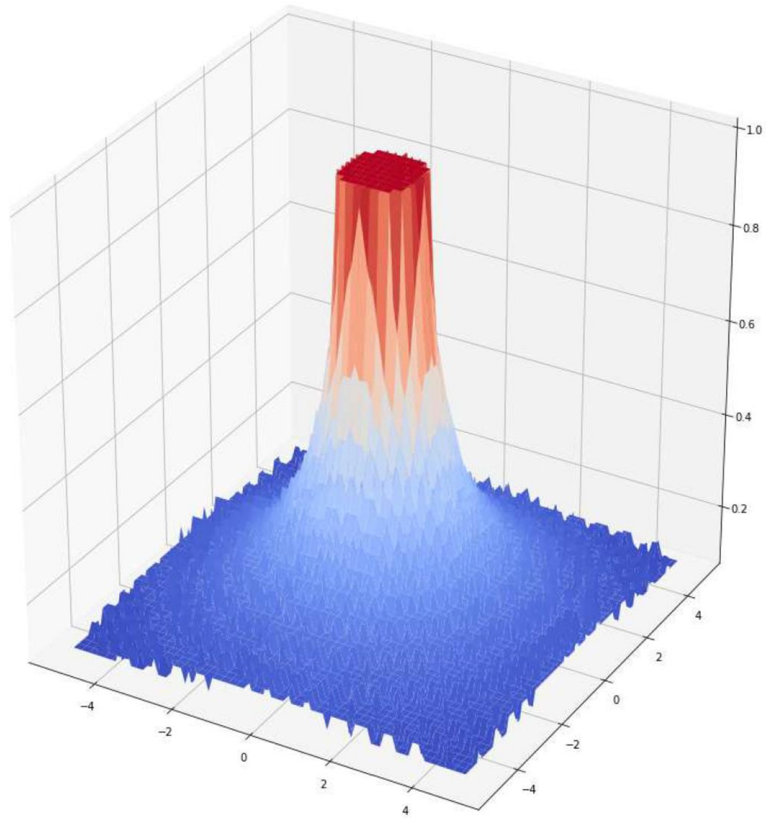
16 sides



- What are the sums in the different regions?
  - A pattern emerges as we consider N > 6..

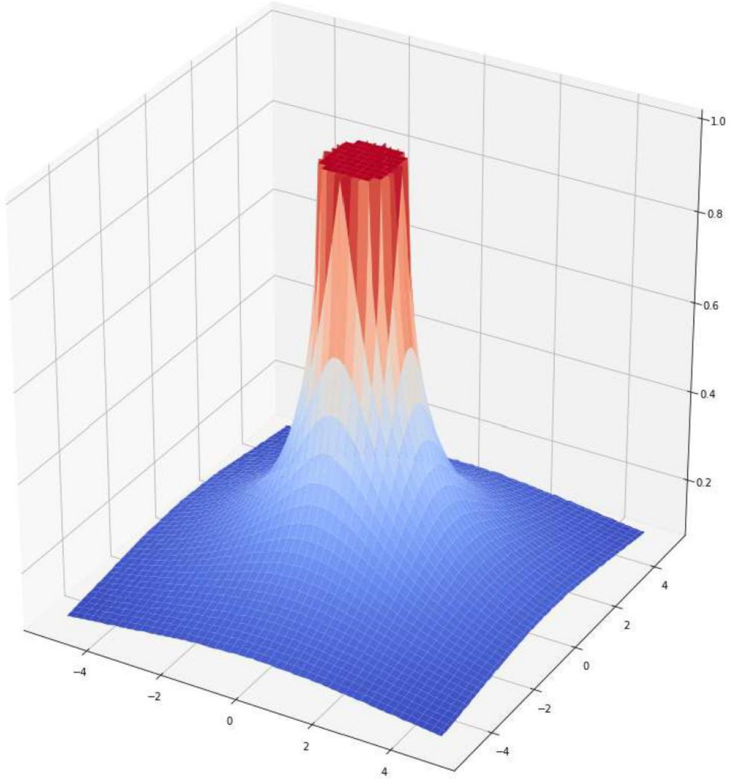
Converging to a Cylinder

64 sides



- What are the sums in the different regions?
  - A pattern emerges as we consider N > 6..

Converging to a Cylinder

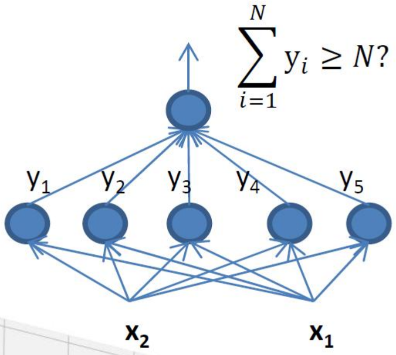


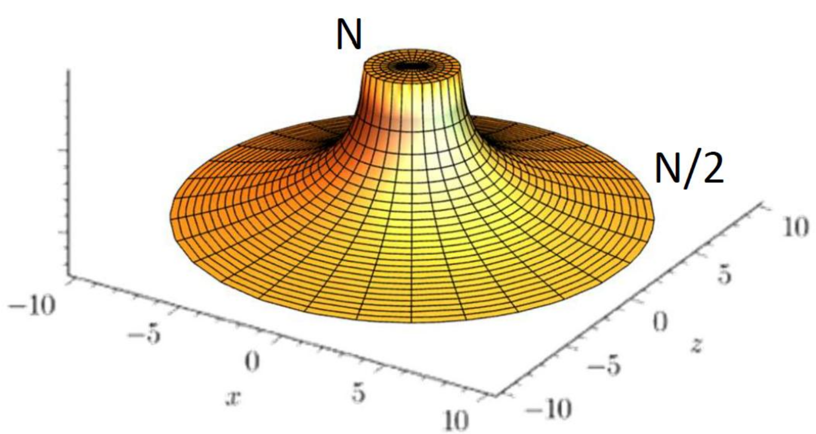
Increasing the number of sides reduces the area outside the polygon that have

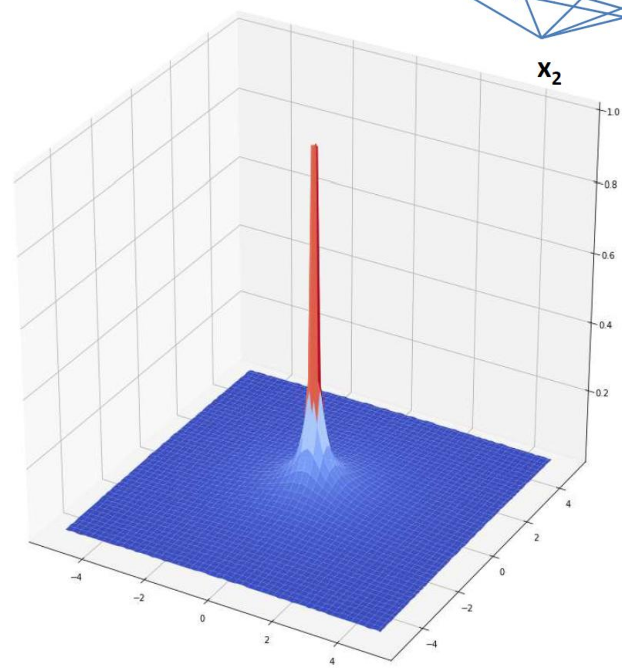
$$\frac{N}{2} < \sum_i y_i < N$$

Converging to a Cylinder

In the limit



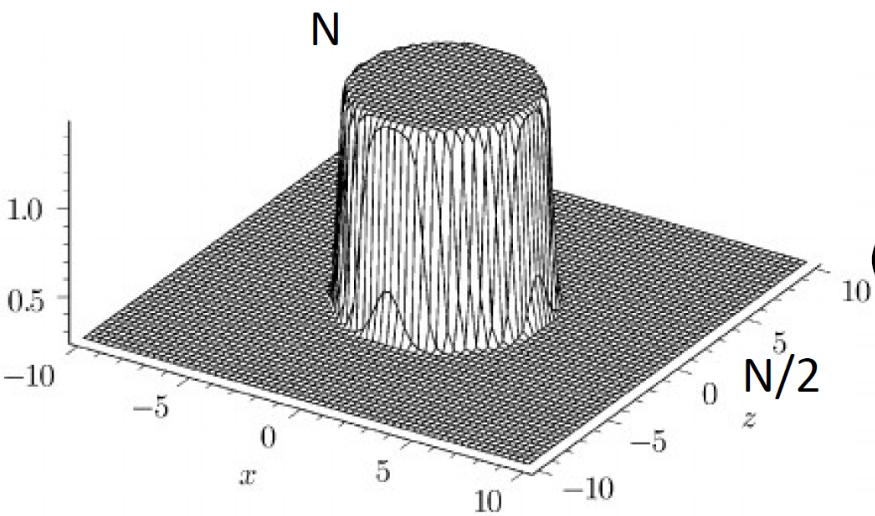


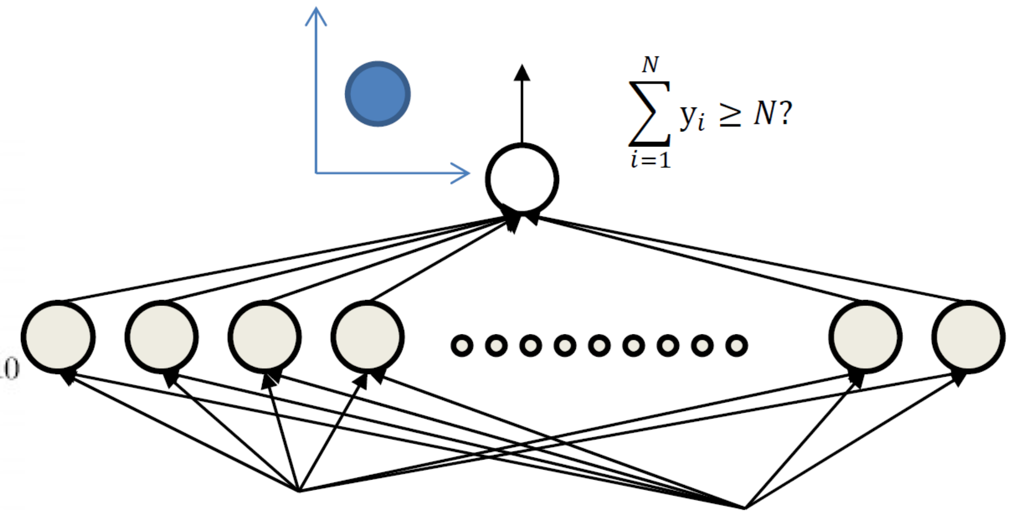


For small radius, it's a near perfect cylinder

– N in the cylinder, N/2 outside

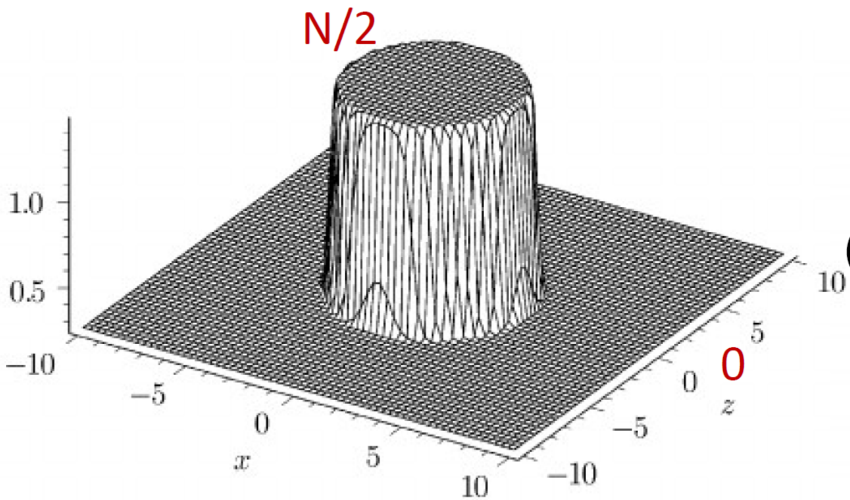
Approximating any Region with Cylinders

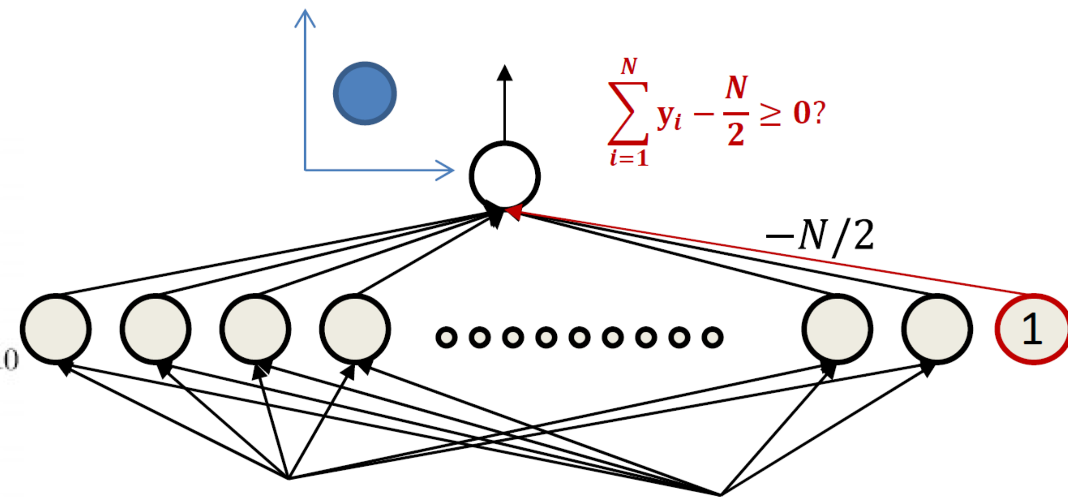




- The circle net
  - Very large number of neurons
  - *Sum is N inside the circle, N/2 outside almost everywhere*
  - Circle can be at any location

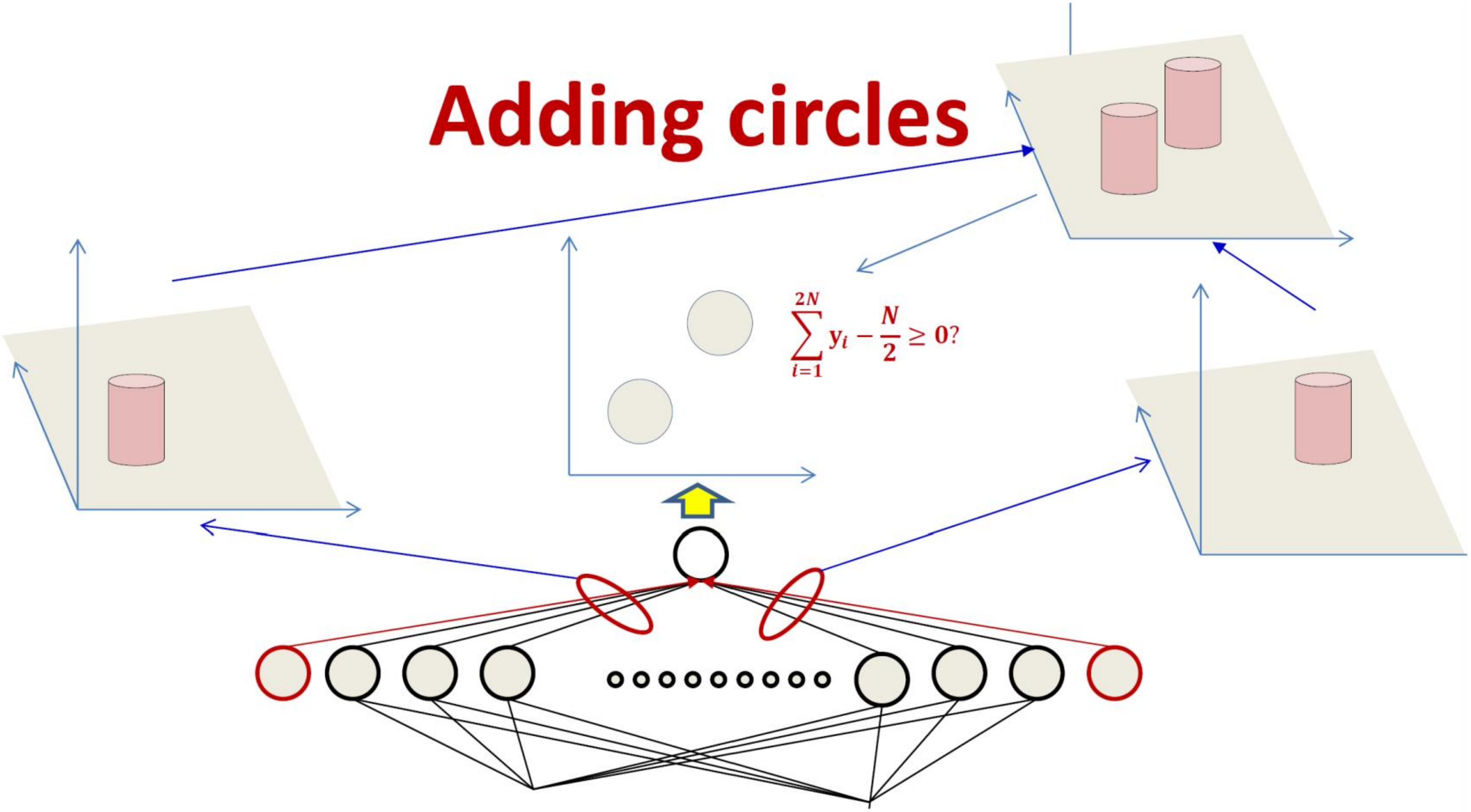
Approximating any Region with Cylinders





- The circle net
  - Very large number of neurons
  - *Sum is N/2 inside the circle, 0 outside almost everywhere*
  - Circle can be at any location

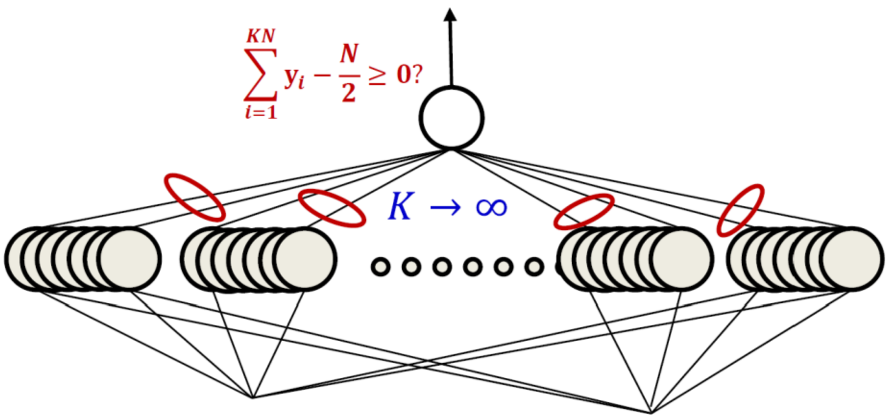
Approximating any Region with Cylinders

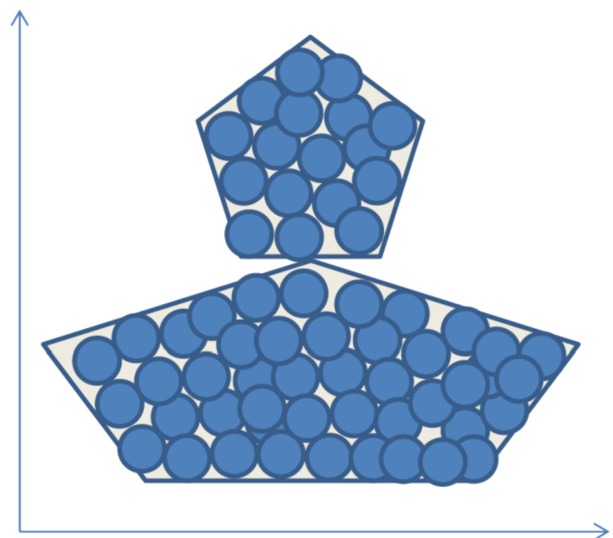


- The "sum" of two circles sub nets is exactly N/2 inside either circle, and 0 almost everywhere outside

Approximating any Region with Cylinders

Composing an arbitrary figure





- Just fit in an arbitrary number of circles
  - More accurate approximation with greater number of smaller circles
  - Can achieve arbitrary precision

Thank you